40. What is meant by a reversible and irreversible processes?
41. State Clausius form of the second law of thermodynamics
42. State Kelvin-Planck statement of second law of thermodynamics.
43. Define heat engine.
44. What are processes involves in a Carnot engine?
45. Can the given heat energy be completely converted to work in a cyclic process? If not, when can the heat can completely converted to work?
46. State the second law of thermodynamics in terms of entropy.
47. Why does heat flow from a hot object to a cold object?
48. Define the coefficient of performance.

## III. Long answer Questions:

1. Explain the meaning of heat and work with suitable examples.
2. Discuss the ideal gas laws.
3. Explain in detail the thermal expansion.
4. Describe the anomalous expansion of water. How is it helpful in our lives?
5. Explain Calorimetry and derive an expression for final temperature when two thermodynamic systems are mixed.
6. Discuss various modes of heat transfer.
7. Explain in detail Newton's law of cooling.
8. Explain Wien's law and why our eyes are sensitive only to visible rays?
9. Discuss the
   a. thermal equilibrium
   b. mechanical equilibrium
   c. Chemical equilibrium
   d. thermodynamic equilibrium.
10. Explain Joule's Experiment of the mechanical equivalent of heat.
11. Derive the expression for the work done in a volume change in a thermodynamic system.
12. Derive Mayer's relation for an ideal gas.
13. Explain in detail the isothermal process.
14. Derive the work done in an isothermal process
15. Explain in detail an adiabatic process.
16. Derive the work done in an adiabatic process
17. Explain the isobaric process and derive the work done in this process
18. Explain in detail the isochoric process.
19. What are the limitations of the first law of thermodynamics?
20. Explain the heat engine and obtain its efficiency.
21. Explain in detail Carnot heat engine.
22. Derive the expression for Carnot engine efficiency.
23. Explain the second law of thermodynamics in terms of entropy.
24. Explain in detail the working of a refrigerator.



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